American Computer Science League

2024-2025 ● Contest 1: Rings ● Senior Division

PROBLEM: ACSL Rings is a game where the object is to score points by tossing a ball through one of the 5 rings. Points are awarded as follows:



- Through the aqua or red ring - 1 point
- Through the orange or green ring - 3 points
- Through the black ring - 6 points
- Through the space where 2 rings overlap - the sum of the points for the 2 rings plus 1
- Any toss that is identified with a "+" at the end of the substring is awarded 2 extra points for having been tossed from a long distance

We will use a single letter to represent each of the above colors: aqua (A), red (R), orange (O), green (G), and black (B). The overlapping areas will be given by 2 letters: aqua and orange (AO or OA), orange and black (OB or BO), black and green (BG or GB), and green and red (GR or RG).

You will be given an integer, *n*, indicating the number of players, followed by *n* separate strings representing each player's results for all tosses that go through at least one ring. Output a string for each player's score in descending order using the format "player#-score", each separated by a single space. Ties are broken based on the number of tosses; the fewer the tosses the better. We guarantee that all ties will be broken.

EXAMPLE:

Input	`3`
	`A R AO+`
	`B+ GR OA B`
	`G OB+`

Output	`2-24 3-15 1-9`
Explanation	Player 1's tossess go through the aqua, red, and aqua & orange rings for a score of 1 + 1 + (1+3+1+2) = 9. Player 2 has a score of (6+2) + (3+1+1) + (3+1+1) + 6 = 24. Player 3 has a score of 3 + (3+6+1+2) = 15. Player 2 is in first place with 24 points; Player 3 is in second place with 15 points; and Player 1 is in third place with 9 points. The output is the string `2-24 3-15 1-9`.

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问题：ACSL Rings 游戏**规则：向** 5 **个圆环投球，球穿过其中一个圆环**即可**得分。计分规则如下：**

- **球穿过蓝色或红色圆环——得** 1 **分**
- **球穿过橘色或绿色圆环——得** 3 **分**
- **球穿过黑色圆环——得** 6 **分**
- **球穿过两**个圆**环**的**重叠区域——得分为两**个圆**环**的得**分之和再加** 1
- **子串末尾带“**+**”**表示**远距离投球——**玩家可额外获得 2 **分**

我们用单个字母来表示上述**每种颜色的圆环：蓝色（**A**）、红色（**R**）、橘色（**O**）、绿色（**G**）、黑色（**B**）。两**个圆**环**的**重叠区域用两个字母来表示：蓝色和橘色圆环重叠区域**（AO 或 OA）、**橘色和黑色圆环重叠区域**（OB 或 BO）、**黑色和绿色圆环重叠区域**（BG 或 GB）、**绿色和红色圆环重叠区域**（GR 或 RG）。

给定**一个整**数 *n* 表示**玩家数量，后跟** *n* 个**独立字符串，分别表示每位玩家投球穿过**至少一个圆环的所有结果。将**每位玩家的得分**按照**“玩家序号-得分”**的格式**降序输出为一个字符串，且每位玩家**的得分**之间用一个空格隔开。如**玩家得分**出现平局，则参考每位玩家的投球次数，投球次数更少的**玩家**优先。我们确定所有平局都可**被**打破。**

示例：

输入	3 A R AO+ B+ GR OA B G OB+
输出	2-24 3-15 1-9
解析	1 号**玩家投球穿过蓝色圆环、红色圆环、蓝色和橘色圆环的重叠区域，得分为** 1 + 1 + (1+3+1+2) = 9 。 2 号**玩家得分为** (6+2) + (3+1+1) + (3+1+1) + 6 = 24 。 3 号**玩家得分为** 3 + (3+6+1+2) = 15 。 2 号**玩家得分为** 24 **分，排第一位；**3 **号玩家得分为** 15 **分，排第二位；**1 **号玩家得分为** 9 **分，排第三位。输出字符串：**2-24 3-15 1-9 。

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INPUT: The input will consist of an integer representing the number of players followed by a string for each player's results as described above.

OUTPUT: Output a string in the format "player#-score" for each player's score in descending order, each separated by a single space.

输入：输入**一个整数，表示玩家数量，后跟字符串，表示每位玩家**的**投球**结果，格式**如上所述。**

输出：将**每位玩家的得分按照“ 玩家序号-得分 ”的格式降序输出为一个字符串，每位玩家**的得分**之间用一个空格隔开。**

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SAMPLE INPUT:

```
1. 3
   A R AO+
   B+ GR OA B
   G OB+
2. 3
   A B R G O
   AO OA BO OB GR RG BG GB
   A AO B OB R GR B GB
3. 5
   A+ B R+ G O+
   AO OB+ GB RG
   A AO B R GR G GB O BO
   B R A G GR A BG OA R
   BG+ B+ G+ OA+ R+ A+ O+
4. 3
   A+ R+ AO+ B+ BG+ R+
   B+ GR+ AO+ B+
   GR+ OB+ R+ AO+ G+ R+
5. 4
   A R+ OA B AO OB BG GR B+
   AO OB B+ G BG BG+ GR+
   B+ G AO+ OB GR O+ B
   B+ B G AO+ AO BG GR G O+ G
6. 6
   R R G G AO BG GB B
   AO GR+ A B AO+ BG BG BO
   A A R R G G O O
   BG GB GR+ RG O O+ O A+ B
   GR RG A+ A O O+ B G
   A B+ B O G+ G G AO OB GR BG
```

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样例输入：

```
1. 3
   A R AO+
   B+ GR OA B
   G OB+
2. 3
   A B R G O
   AO OA BO OB GR RG BG GB
   A AO B OB R GR B GB
3. 5
   A+ B R+ G O+
   AO OB+ GB RG
   A AO B R GR G GB O BO
   B R A G GR A BG OA R
   BG+ B+ G+ OA+ R+ A+ O+
4. 3
   A+ R+ AO+ B+ BG+ R+
   B+ GR+ AO+ B+
   GR+ OB+ R+ AO+ G+ R+
5. 4
   A R+ OA B AO OB BG GR B+
   AO OB B+ G BG BG+ GR+
   B+ G AO+ OB GR O+ B
   B+ B G AO+ AO BG GR G O+ G
6. 6
   R R G G AO BG GB B
   AO GR+ A B AO+ BG BG BO
   A A R R G G O O
   BG GB GR+ RG O O+ O A+ B
   GR RG A+ A O O+ B G
   A B+ B O G+ G G AO OB GR BG
```

SAMPLE OUTPUT:

```
1. 2-24 3-15 1-9
2. 2-60 3-44 1-14
3. 3-44 5-43 4-33 2-32 1-20
4. 3-37 1-36 2-30
5. 2-55 4-55 1-53 3-44
6. 6-59 2-56 4-52 1-39 5-31 3-16
```

样例输出：

1. 2-24 3-15 1-9
2. 2-60 3-44 1-14
3. 3-44 5-43 4-33 2-32 1-20
4. 3-37 1-36 2-30
5. 2-55 4-55 1-53 3-44
6. 6-59 2-56 4-52 1-39 5-31 3-16

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TEST DATA

TEST INPUT:

1. 3
   G R G R B G B G O B O B O B A O A O A O A O
   B B B G G A A A O B A O B G B G A O A O
   G B O A A A A B B B O O O G G G A O O B G R B G
2. 4
   GR RG BG GB OB BO OA AO
   G R O B A GB OA RG OB
   AO
   BG
3. 5
   A O B+ GR OB R AO+ G R GR+ BG
   AO GR BG+ A R O+ G B AO OB BG GR+ O R R+ B
   BG AO+ BG A A B B G AO AO+ BG GR B G A O A
   AO OB R GR+ B G R AO OB GR O B+ O A B
   B+ O G AO OB B+ G A+ B BG GR A O G R
4. 6
   B+ B+ R A O GR+ GR BG BG+ OB OB OB AO AO AO AO+ A R G B B
   AO BG GR GR B B G G+ G A A A GR OB AO BG BG AO+ AO
   A B G+ O R R GR BG OB AO A B B B O O G AO OB+ GR BG B
   GR AO BG A O B+ AO GR BG BG OB+ OB OB AO AO AO AO O G B A
   GR BG AO OB O+ O O B G A A GR OB AO BG+ AO G B B R
   A A+ R R BG G G O O+ O A A R R B B B BG+ G O O
5. 6
   GR AO BG R R G+ G AO AO+ BG BG B R R A A G
   B A O R+ G G AO GR A B R G O BG BG OB B
   BG BG AO GR A A A B B R R G G G O O O O AO BG GR
   A A A B B+ B O O+ O G G G AO OB GR BG O R B O
   R B B G A BG BG GR GR A A A O+ O O B B G
   A R G B+ BG BG+ GR GR O O O A B AO OB BG
6. 7
   A R B G+ O B G O+ A R B+ G AO AO BG GR GR BG OB OB OB AO
   B G O R A AO BG GR OB AO BG GR OB AO BG GR OB AO BG GR+ OB
   B R A G GR A B R OB+ B A R B OB R A G BG+ G AO OB OB OB BG
   OB+ GB+ GR BG+ RG+ BG+ OA+ OB+ G+ B+ GB+ BO+ AO BO+ G
   AO GR GR OB+ OB OB AO GR GR+ OB OB OB BG BG BG BG B R A G O
   AO OB+ GR BG+ OB GR+ OB GR+ BG AO+ OB GR+ BG+ R+ A+ G O+ B+
   BG OA+ GR B+ B R+ RG A+ G G AO O B+ A G GB OB+ BO OA+ O G

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测试数据

测试输入：

1. 3
   G R G R B G B G O B O B O B A O A O A O A O
   B B B G G A A A O B A O B G B G A O A O
   G B O A A A A B B B O O O G G G A O O B G R B G
2. 4
   GR RG BG GB OB BO OA AO
   G R O B A GB OA RG OB
   AO
   BG
3. 5
   A O B+ GR OB R AO+ G R GR+ BG
   AO GR BG+ A R O+ G B AO OB BG GR+ O R R+ B
   BG AO+ BG A A B B G AO AO+ BG GR B G A O A
   AO OB R GR+ B G R AO OB GR O B+ O A B
   B+ O G AO OB B+ G A+ B BG GR A O G R
4. 6
   B+ B+ R A O GR+ GR BG BG+ OB OB OB AO AO AO AO+ A R G B B
   AO BG GR GR B B G G+ G A A A GR OB AO BG BG AO+ AO
   A B G+ O R R GR BG OB AO A B B B O O G AO OB+ GR BG B
   GR AO BG A O B+ AO GR BG BG OB+ OB OB AO AO AO AO O G B A
   GR BG AO OB O+ O O B G A A GR OB AO BG+ AO G B B R
   A A+ R R BG G G O O+ O A A R R B B B BG+ G O O
5. 6
   GR AO BG R R G+ G AO AO+ BG BG B R R A A G
   B A O R+ G G AO GR A B R G O BG BG OB B
   BG BG AO GR A A A B B R R G G G O O O O AO BG GR
   A A A B B+ B O O+ O G G G AO OB GR BG O R B O
   R B B G A BG BG GR GR A A A O+ O O B B G
   A R G B+ BG BG+ GR GR O O O A B AO OB BG
6. 7
   A R B G+ O B G O+ A R B+ G AO AO BG GR GR BG OB OB OB AO
   B G O R A AO BG GR OB AO BG GR OB AO BG GR OB AO BG GR+ OB
   B R A G GR A B R OB+ B A R B OB R A G BG+ G AO OB OB OB BG
   OB+ GB+ GR BG+ RG+ BG+ OA+ OB+ G+ B+ GB+ BO+ AO BO+ G
   AO GR GR OB+ OB OB AO GR GR+ OB OB OB BG BG BG BG B R A G O
   AO OB+ GR BG+ OB GR+ OB GR+ BG AO+ OB GR+ BG+ R+ A+ G O+ B+
   BG OA+ GR B+ B R+ RG A+ G G AO O B+ A G GB OB+ BO OA+ O G

TEST OUTPUT:

1. 3-78 1-69 2-66
2. 1-60 2-44 4-10 3-5
3. 3-85 2-83 4-74 5-72 1-56
4. 4-127 1-124 3-113 5-105 2-103 6-76
5. 3-88 6-86 4-86 2-79 5-76 1-75
6. 5-148 4-136 6-136 2-136 3-125 7-118 1-118

测试输出：

1. 3-78 1-69 2-66
2. 1-60 2-44 4-10 3-5
3. 3-85 2-83 4-74 5-72 1-56
4. 4-127 1-124 3-113 5-105 2-103 6-76
5. 3-88 6-86 4-86 2-79 5-76 1-75
6. 5-148 4-136 6-136 2-136 3-125 7-118 1-118